

(6.3)

input-state linearization of SISO systems

single input nonlinear system

$$\dot{x} = f(x) + g(x)u, \quad \text{---} (*)$$

where $f, g : \mathbb{C}^\infty$ -vector fields

→ affine in control

Def (*) is said to be input-state linearizable if

$\exists$ a region $\Omega \subset \mathbb{R}^n$, a diffeo $\phi: \Omega \rightarrow \mathbb{R}^n$, a nonlinear feedback control law

$$u = \alpha(x) + \beta(x)v$$

s.t. the new state variable $z = \phi(x)$, the new input v satisfy

$$\dot{z} = Az + bv, \quad \text{---} (**)$$

where $A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$

$$\begin{aligned} \dot{w} &= Bw + cv, \\ \dot{w}_i &= \sum_{j=1}^n b_{ij} w_j + c_i v, \\ z &= (w, w_1, \dots, w_{n-1})^T \\ w_1^{(n)} &= \bigcirc + \bigcirc \cdot \bigcirc \cdot \bigcirc \\ v &= \frac{\bigcirc}{c} \end{aligned}$$

* z : linearizing state, $u = \alpha(x) + \beta(x)v$: linearizing control law

* special form of A, b does not lose generality.

(*) special case of input-output linearization with relative degree n .

(if we put the output function to be z_1)

the converse is also true. $y^{(n)} = v \rightarrow z = (y, y', \dots, y^{(n-1)}) \Rightarrow$ input-state lin.

Lemma 6.3 n -th-order nonlinear system : input-state linearizable

$\Leftrightarrow \exists$ scalar function $z(x)$ s.t. input-output linearization with $z(x)$ as output function has relative degree n .

Thm 6.2 $(\dot{x} = f(x) + g(x)u)$ (f) : input-state linearizable

$\Leftrightarrow \exists$ region Ω s.t. ① vector fields $\{g, \text{ad}_f g, \dots, \text{ad}_f^{n-1} g\}$ are linearly independent in Ω

② $\{g, \text{ad}_f g, \dots, \text{ad}_f^{n-2} g\}$ involutive in Ω

* ① : controllability, ② : involutivity

Lemma 6.4 $z(x) : C^\infty$ -fun in a region Ω . In Ω ,

$$L_g z = L_g L_f z = \dots = L_g L_f^k z = 0$$

$$\Leftrightarrow L_g z = L_{\text{ad}_f g} z = \dots = L_{\text{ad}_f^k g} z = 0$$

Pf, $(\Rightarrow)$ $k=0$: $\checkmark$

$$k=1 : L_{\text{ad}_f g} z = L_f L_g z - L_g L_f z = 0 - 0 = 0$$

$$k=2 : L_{\text{ad}_f^2 g} z = (L_f L_{\text{ad}_f g} - L_{\text{ad}_f g} L_f) z = (L_f^2 L_g - 2L_f L_g L_f + L_g L_f^2) z$$

inductively ...

$(\Leftarrow)$ Similar.

pf of thm 6.2)

$(\Rightarrow) \exists$ state transformation $\bar{z} = z(x)$,

input $u = \alpha(x) + \beta(x)v$ s.t. $\dot{\bar{z}} = A\bar{z} + bv$

$$\stackrel{?}{\Leftrightarrow} \left(\{g, \text{ad}_f g, \dots, \text{ad}_f^{n-1} g\} : \text{linearly independent} \right)$$

L_f : Lie derivative

$$L_f z = \nabla z \cdot f$$

$$\left(\{g, \text{ad}_f g, \dots, \text{ad}_{f^{n-2}} g\} : \text{inductive} \right)$$

$$\ddot{z}_1 = \frac{\partial z_1}{\partial x} \cdot (f + g u) = z_2 \quad (\text{compute the 1st coordinate of } \ddot{z} = A\ddot{z} + b\ddot{v})$$

Similarly, we obtain

$$\begin{aligned} \frac{\partial z_1}{\partial x} f + \frac{\partial z_1}{\partial x} g u &= z_2 \\ \frac{\partial z_2}{\partial x} f + \frac{\partial z_2}{\partial x} g u &= z_3 \end{aligned}$$

$$\vdots$$

$$\frac{\partial z_n}{\partial x} f + \frac{\partial z_n}{\partial x} g u = v$$

$z_1, \dots, z_n$: independent of u , v is not

$$\Rightarrow \begin{cases} L_g z_1 = \dots = L_g z_{n-1} = 0, \quad L_g z_n \neq 0, \\ L_f z_i = z_{i+1}, \quad i=1, 2, \dots, n-1 \end{cases}$$

$$L_g z_1 = L_g L_f z_1 = \dots = L_g L_f^{n-2} z_1 = 0$$

$$\Rightarrow L_g z_1 = L_{\text{ad}_f g} z_1 = \dots = L_{\text{ad}_f^{n-2} g} z_1 = 0 \quad \text{by Lem 6.4.}$$

$$\nabla z_1 \cdot g = \nabla z_1 \cdot \text{ad}_f g = \dots = \nabla z_1 \cdot \text{ad}_f^{n-2} g = 0 \quad \text{--- (*)}$$

$$L_{\text{ad}_f g} = L_f L_g - L_g L_f$$

$$\begin{aligned} L_{\text{ad}_f^2 g} &= L_f (L_f L_g - L_g L_f) - (L_f L_g - L_g L_f) L_f \\ &= L_f^2 L_g - 2L_f L_g L_f + L_g L_f^2 \end{aligned}$$

$$L_{\text{ad}_f^3 g} = L_f (L_f^2 L_g - 2L_f L_g L_f + L_g L_f^2) - (L_f^2 L_g - 2L_f L_g L_f + L_g L_f^2) L_f$$

$$= \dots - L_g L_f^3$$

⋮

$$L_{\text{ad}_f^{n-1}g} = \dots + (-1)^{n-1} L_g L_f^{n-1}$$

$$\Rightarrow \nabla z_i \cdot \text{ad}_f^{n-1}g = L_{\text{ad}_f^{n-1}g} z_i = (-1)^{n-1} L_g L_f^{n-1} z_i = (-1)^{n-1} L_g z_n \neq 0 \quad (*)2$$

From $(*)1, (*)2$, $\{g, \text{ad}_f g, \dots, \text{ad}_f^{n-1}g\}$: linearly independent

∴ If not, $\text{ad}_f^i g = \sum_{j=0}^{i-1} \alpha_j \text{ad}_f^j g$, α_j : scalar function

$$\Rightarrow \text{ad}_f^{n-1}g = \sum_{j=n-1}^{n-2} \alpha_{j-n+1} \text{ad}_f^j g$$

← Pick the largest i
i.e. $\alpha_i \neq 0$.

$$\Rightarrow \nabla z_i \cdot \text{ad}_f^{n-1}g = 0, \text{ a contradiction.}$$

From $(*)1$, $\{g, \text{ad}_f g, \dots, \text{ad}_f^{n-2}g\}$ is completely integrable, and hence involutive by Frobenius theorem.

($\Leftarrow$) $\{g, \text{ad}_f g, \dots, \text{ad}_f^{n-1}g\}$: linearly independent

$\{g, \text{ad}_f g, \dots, \text{ad}_f^{n-2}g\}$: involutive

(?)
 $\Rightarrow$ we can find state transf, input transf

$$\exists \text{ scalar fn } z_i(x) \text{ s.t. } \nabla z_i \neq 0, L_g z_i = L_{\text{ad}_f g} z_i = \dots = L_{\text{ad}_f^{n-2}g} z_i = 0$$

(Lem 6.4)

$$\Rightarrow L_g z_i = L_g L_f z_i = \dots = L_g L_f^{n-2} z_i = 0$$

$$z = (z_1, L_f z_1, \dots, L_f^{n-1} z_1)^T \text{ new set of state variables}$$

$$\rightarrow \begin{cases} \dot{z}_k = z_{k+1}, & k=1, \dots, n-1, \\ \dot{z}_n = L_f^{n-1} z_1 + L_g L_f^{n-1} z_1 u \end{cases}$$

$$L_g L_f^{n-r} z_1 = (-1)^{n-r} L_{ad_f^{n-r} g} z_1$$

$$L_{ad_f^{n-r} g} z_1(x) \neq 0, \quad \forall x \in \Omega. \quad \text{Otherwise,}$$

$$\nabla z_1 \cdot (g \quad ad_f g \quad \dots \quad ad_f^{n-r} g) = 0$$

$\Rightarrow \nabla z_1$: orthogonal to n linearly independent vectors

$\Rightarrow \nabla z_1(x) = 0$, a contradiction

Take the control law $u = (-L_f^n z_1 + v) / L_g L_f^{n-r} z_1$

$$\Rightarrow \dot{z}_1 = v$$

$\therefore$ input-state linearization achieved!

$$\begin{aligned} \dot{z}_1 &= \frac{\partial z_1}{\partial x} \dot{x} = \nabla z_1 \cdot \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} \\ z_1 &= z_1(x) = \nabla z_1 \cdot (f(x) - g(x)u) \\ \dot{x} &= Affine system. \quad || \\ L_f z_1 &\neq L_g z_1 u \end{aligned}$$

Strategy for input-state linearization

- ① Construct the vector fields $g, \text{ad}_f g, \dots, \text{ad}_f^{n-1} g$
- ② Check the controllability and involutivity conditions
- ③ If ② satisfied, find the first state z_1 from equations

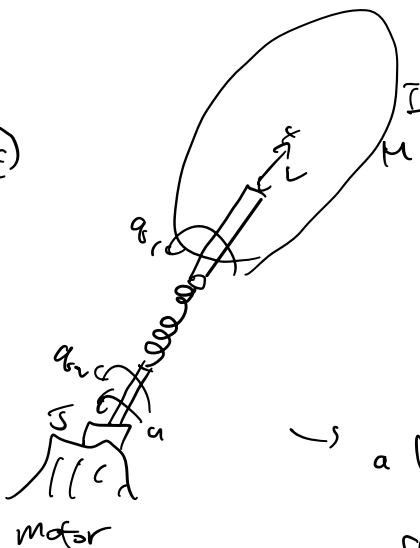
$$\begin{cases} \nabla_{z_1} \text{ad}_f^i g = 0, & i=0, \dots, n-2, \\ \nabla_{z_1} \text{ad}_f^{n-1} g \neq 0 \end{cases}$$

④ state transformation $z(x) = [z_1 \ L_f z_1 \ \dots \ L_f^{n-1} z_1]^T,$

input transformation $u = \alpha(x) + \beta(x)v$ with

$$\alpha(x) = - \frac{L_f^n z_1}{L_g L_f^{n-1} z_1}, \quad \beta(x) = \frac{1}{L_g L_f^{n-1} z_1}$$

(ex)



$$I \ddot{q}_1 + m g L \sin q_1 + k(q_1 - q_2) = 0$$

$$J \dot{q}_2 - k(q_1 - q_2) = u$$

↪ a link driven by a motor through a torsional spring flexible-joint mechanism.

$$x = (q_1 \ \dot{q}_1 \ q_2 \ \dot{q}_2)^T \quad \text{state vector}$$

$$\begin{cases} f = \begin{pmatrix} \dot{x}_2 \\ -\frac{m g L}{I} \sin x_1 - \frac{k}{I}(x_1 - x_3) \\ 0 \\ 0 \end{pmatrix} & g = \begin{pmatrix} 0 \\ 0 \\ \frac{k}{J}(x_1 - x_3) \\ \frac{1}{J} \end{pmatrix}^T \end{cases}$$

Let's check the controllability and the redundancy conditions.

controllability matrix :

$$(g \quad \text{ad}_f g \quad \text{ad}_f^2 g \quad \text{ad}_f^3 g) = \begin{pmatrix} 0 & 0 & 0 & -\frac{k}{IJ} \\ 0 & 0 & \frac{k}{IJ} & 0 \\ 0 & -\frac{1}{J} & 0 & \frac{k}{J} \\ \frac{1}{J} & 0 & -\frac{k}{J^2} & 0 \end{pmatrix}$$

* $\text{ad}_f g = [f, g]$
 $= (\nabla_{g_i} f - \nabla_{f_i} g)_i$

$\rightarrow$ rank 4 for $k > 0$, $IJ < \infty$

$(g \quad \text{ad}_f g \quad \text{ad}_f^2 g)$: constant vector fields $\rightarrow$ involutive ($\because$ Lie bracket = 0)

$\therefore$ the system is input-state linearizable.

the first component z_1 of z satisfies

* $L_g z_1 = 0, L_{\text{ad}_f g} z_1 = 0,$
 $L_{\text{ad}_f^2 g} z_1 = 0, L_{\text{ad}_f^3 g} z_1 \neq 0$

$$\frac{\partial z_1}{\partial x_2} = 0 \quad \frac{\partial z_1}{\partial x_3} = 0 \quad \frac{\partial z_1}{\partial x_f} = 0 \quad \frac{\partial z_1}{\partial x_r} \neq 0$$

from $\nabla z_1 \cdot g = \nabla z_1 \cdot \text{ad}_f g = \nabla z_1 \cdot \text{ad}_f^2 g = 0$, $\nabla z_1 \cdot \text{ad}_f^3 g \neq 0$

simple solution : $z_1 = x_r$

$\rightarrow z_2 = \nabla z_1 \cdot f = x_r$

$$z_3 = \nabla z_2 \cdot f = -\frac{MgL}{I} \sin x_r - \frac{k}{I} (x_r - x_f)$$

$$z_f = \nabla z_3 \cdot f = -\frac{MgL}{I} x_r \cos x_r - \frac{k}{I} (x_r - x_f)$$

$$u = \frac{-L_f^3 z_1 + v}{L_g L_f^3 z_1} = \frac{v - \nabla z_f \cdot f}{\nabla z_f \cdot g} = \frac{v - a(x)}{\frac{k}{IJ}}$$

where $a(x) = \frac{MgL}{I} x_r^2 \sin x_r + \left(\frac{MgL}{I} \sin x_r + \frac{k}{I} (x_r - x_f) \right) \left(\frac{MgL}{I} \cos x_r + \frac{k}{I} \right) + \frac{k^2}{IJ} (x_r - x_f)$

$$\Rightarrow \begin{cases} \dot{z}_1 = z_2 \\ \dot{z}_2 = z_3 \\ \dot{z}_3 = z_4 \\ \dot{z}_4 = v \end{cases}$$

Note

• above input-state linearization is global since $z = z(x)$, input transformation is well-defined everywhere

• transformed variables z_1 : link position
 z_2 : link velocity
 z_3 : link acceleration
 z_4 : link jerk

• $\nabla z_1 \cdot \text{ad}_f^{n-1} g \neq 0 \rightarrow \nabla z_1 \cdot \text{ad}_f^{n-1} g = 1$
 normalization

$$\Rightarrow (\text{ad}_f^0 g \quad \text{ad}_f^1 g \quad \dots \quad \text{ad}_f^{n-1} g) \begin{pmatrix} \frac{\partial z_1}{\partial x_1} \\ \vdots \\ \frac{\partial z_1}{\partial x_n} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

$\frac{\partial z_1}{\partial x_1}, \dots, \frac{\partial z_1}{\partial x_n}$ uniquely determined,

z_1 can be found by sequentially integrating these partial derivatives.

After state equation transformed into a linear form, one can design controllers for either stabilization or tracking.

$$z_r^{(4)} = v$$

If we want z_r to track $z_{dr}(t)$, $v = z_{dr}^{(4)} - a_3 \tilde{z}_r^{(3)} - a_2 \tilde{z}_r^{(2)} - a_1 \tilde{z}_r^{(1)} - a_0 \tilde{z}_r$

$$(\tilde{z}_r = z_r - z_{dr})$$

$$\rightarrow \tilde{z}_r^{(4)} + a_3 \tilde{z}_r^{(3)} + \dots + a_0 \tilde{z}_r = 0$$

$\rightarrow$ exponentially stable if $p^4 + a_3 p^3 + \dots + a_0 = 0$ has all roots in left complex plane.

§ 6.4. Input-Output Linearization (of SISO Systems)

$$\begin{cases} \dot{x} = f(x) + g(x)u \\ y = h(x) \end{cases} \rightarrow \text{system output}$$

Goal: generation of a linear differential relation betw y and v . ↗ new input

- How to generate?
- internal dyn, zero-dyn?
- How to design state controllers?

in a region Ω_x in the state space,

$$\dot{y} = Dh(f+gu) = L_f h(x) + L_g h(x)u$$

If $L_g h(x) \neq 0$, $x_0 \in \Omega_x \Rightarrow L_g h(x) \neq 0$, for x in a neighborhood of x_0

$$\text{Put } u = \frac{1}{L_g h} (-L_f h + v) \text{ in } \Omega$$

$$\Rightarrow \dot{y} = v$$

If $L_g h(x) = 0$, $\forall x \in \Omega_x$, $\dot{y} = L_f h(x)$ $\rightarrow$ differentiate $\dot{y} \Rightarrow$

$$\ddot{y} = L_f^2 h(x) + L_g L_f h(x)u$$

$$y^{(3)} = L_f^3 h(x) + L_g L_f^2 h(x)u$$

If $L_g L_f h(x) = 0$, $\forall x \in \Omega_x$, $\ddot{y} = L_f^2 h(x) \rightarrow$ diff again $\rightarrow \dots$

until $L_g L_f^{r-1} h(x) \neq 0 \exists x_0 \text{ in } \Omega_x \Rightarrow L_g L_f^{r-1} h(x) \neq 0$ for x in a neighborhood of x_0

$$\text{Put } u = \frac{1}{L_g L_f^{r-1} h} (-L_f^r h + v) \text{ in } \Omega$$

$$L_g h(x) = L_g L_f h(x) = \dots = L_g L_f^{r-2} h(x) = 0$$

$$\Rightarrow y^{(r)} = v$$

$y = h(x), \dot{y} = L_f h(x), \ddot{y} = L_f^2 h(x), \dots, y^{(r-1)} = L_f^{r-1} h(x), y^{(r)} = L_f^r h(x) + L_g L_f^{r-1} h(x)u$

* r : rel degree, $r \leq n$, $r = n$: input-state linearization in Ω 4

Def SISO system has relative degree r in Ω if

$$\forall x \in \Omega, \quad L_g L_f^i h(x) = 0, \quad 0 \leq i < r-1, \quad L_g L_f^{r-1} h(x) \neq 0$$

Normal form

rel deg = $r \rightarrow$ the system transformed into "normal form"

using $y, \dot{y}, \dots, y^{(r-1)}$ as new state variables

$$\mu = [\mu_1 \dots \mu_r]^T = [y \ \dot{y} \ \dots \ y^{(r-1)}]^T$$

normal form of the system: $\dot{\mu} = \begin{bmatrix} \mu_2 \\ \vdots \\ \mu_r \\ a(\mu, \varphi) + b(\mu, \varphi)u \end{bmatrix}$

$\varphi = \omega(\mu, \varphi)$ ↪ $y^{(r)} = L_f^r h(x) + L_g L_f^{r-1} h(x)u$
↪ doesn't contain u

with output $y = \mu_1$

μ_i, φ_j : normal coordinates, normal states

Goal: find diffeo $\phi(x) = [\mu_1, \mu_2, \dots, \mu_r, \varphi_1, \dots, \varphi_{n-r}]^T$ near x_0

$\Leftrightarrow \nabla \mu_1, \dots, \nabla \mu_r, \nabla \varphi_1, \dots, \nabla \varphi_{n-r}$ lin indep.

Lemma 6.5 rel deg r in $\Omega \Rightarrow \nabla \mu_1, \dots, \nabla \mu_r$ lin ind in Ω

pf) $\nabla \mu_i g = 0, 1 \leq i < r, \nabla \mu_r g \neq 0$

$$(r=3) \quad \alpha_1 \nabla \mu_1 + \alpha_2 \nabla \mu_2 + \alpha_3 \nabla \mu_3 = 0$$

$$\rightarrow [\alpha_1 \nabla \mu_1 + \alpha_2 \nabla \mu_2 - \alpha_3 \nabla \mu_3] g = 0$$

$$\Rightarrow \alpha_3 \equiv 0 \text{ in } \Omega$$

$$\alpha_1 \nabla \mu_1 + \alpha_2 \nabla \mu_2 = 0$$

$\rightarrow$ Multiply $\text{ad}_f g$

$$0 = \alpha_1 L_{\text{ad}_f g} \mu_1 + \alpha_2 L_{\text{ad}_f g} \mu_2$$

$$= \alpha_1 (L_f L_g h - L_g L_f h) + \alpha_2 (L_f L_g h - L_g L_f h) L_f h$$

$$= -\alpha_2 \underbrace{L_g L_f^2 h}_{\neq 0}$$

$$\Rightarrow \alpha_2 \equiv 0 \text{ in } \Omega$$

$$\alpha_1 \nabla \mu_1 = 0$$

$\rightarrow$ Multiply $\text{ad}_f^2 g$

$$0 = \alpha_1 L_{\text{ad}_f^2 g} \mu_1 = \alpha_1 [L_f^2 L_g h - 2L_f L_g L_f h + L_g L_f^2 h] = \alpha_1 \underbrace{L_g L_f^2 h}_{\neq 0}$$

$$\Rightarrow \alpha_1 \equiv 0 \text{ in } \Omega$$

$\square$

Cor (rel degree of n th-order system) $\leq n$

pf of $\exists \mathcal{X}_j$) $\nabla \mu_i$ ($i = (n-r-1)$) $\perp$ ortho to g .
is in hyperplane

$\Rightarrow$ can find $(n-1) - (r-1) = n-r$ lin indep vectors $\nabla \mathcal{X}_j$ ($j = (1 \sim n-r)$) $\perp g$

($\because \{g\}$: no solution $\Rightarrow$ $\exists n-1$ lin ind gradient vectors ∇h_k with $\nabla_{h_k} g = 0$)

μ_r : not ortho to g .

$\rightarrow \nabla \mu_i$ ($i = (1 \sim r)$), $\nabla \mathcal{X}_j$ ($j = (1 \sim n-r)$) : lin indep.

$\Rightarrow \phi(x) = (\mu_1 \dots \mu_r \mathcal{X}_1 \dots \mathcal{X}_{n-r})^T$ Jacobian row

$\Rightarrow \phi$: diff'ed in a wtd of x

$$\tilde{\mu} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_r \\ a(\mu, \mathcal{X}) + b(\mu, \mathcal{X})u \end{pmatrix}$$

$$\tilde{\mathcal{X}} = \omega(\mu, \mathcal{X})$$

$$\mu_r^0 = y^{(r)} = L_f^r h(x) + L_g L_f^{r-1} h(x)u = L_f^r h(\phi^T(\mu, \mathcal{X})) + L_g L_f^{r-1} h(\phi^T(\mu, \mathcal{X}))u$$

$$\nabla \mathcal{X}_j \perp g \Rightarrow L_g \mathcal{X}_j = 0 \Rightarrow \tilde{\mathcal{X}} \text{ doesn't contain } u.$$



practical step of finding $\mathcal{X}$.

($\in \mu_1 \dots \mu_r \mathcal{X}_1 \dots \mathcal{X}_{n-r}$ lin indep)

□

$$\textcircled{24} \quad \dot{x} = \begin{pmatrix} -x_1 \\ 2x_1x_2 + \sin x_2 \\ 2x_2 \end{pmatrix} + \begin{pmatrix} e^{2x_2} \\ \frac{1}{2} \\ 0 \end{pmatrix} u$$

$$y = h(x) = x_3$$

$$\rightarrow \dot{y} = 2x_2$$

$$\ddot{y} = 2\dot{x}_2 = 2(2x_1x_2 + \sin x_2) + u$$

$\rightarrow$ relative degree : 2 ,

$$L_f h(x) = 2x_2, \quad L_g h(x) = 0, \quad L_g L_f h(x) = 1$$

normal form

$$\mu_1 = h(x) = x_3$$

$$\mu_2 = L_f h(x) = 2x_2$$

$$\textcircled{\text{Find } \varphi} \quad L_g \varphi = \frac{\partial \varphi}{\partial x_1} e^{2x_2} + \frac{1}{2} \frac{\partial \varphi}{\partial x_2} = 0$$

$$\varphi(x) = 1 + x_1 - e^{2x_2}$$

$$z = [\mu_1 \quad \mu_2 \quad \varphi]^T$$

$$\rightarrow \frac{\partial z}{\partial x} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & -2e^{2x_2} & 0 \end{bmatrix}$$

non-singular for $\forall x$.

inverse given by

$$\begin{cases} x_1 = -1 + \varphi + e^{\mu_2} \\ x_2 = \frac{1}{2}\mu_2 \\ x_3 = \mu_1 \end{cases}$$

$$(x, y, z) \mapsto (\mu_1, \mu_2, \varphi) \text{ global diffeo}$$

The system

$$\begin{cases} \dot{\mu}_1 = \mu_2 \\ \dot{\mu}_2 = 2(1+\mu_2 e^{\mu_2}) + 2\sinh(\mu_2/2) + u \\ \ddot{\varphi} = C(1-\varphi - e^{\mu_2})C(1+2\mu_2 e^{\mu_2}) - 2\sinh(\mu_2/2)e^{\mu_2} \end{cases}$$

) external
) internal

$$\begin{pmatrix} \dot{\varphi} = L_f \varphi(x) - 2e^{2x_2} (2x_1 x_2 + \sinh x_2 + \cancel{\frac{C}{2}u}) \\ = -x_1 - 4x_1 x_2 e^{2x_2} - 2\sinh x_2 e^{2x_2} \end{pmatrix}$$

zero-dynamics ? control input u such that output $y \equiv 0$.

internal dynamics. $\rightarrow$ consider when the output y maintained at 0.

$$\rightarrow y, \dot{y}, \dots, y^{(n-1)} = 0$$

$\rightarrow$ 0 dynamics describes motion on $(n-r)$ -dim C^∞ surface

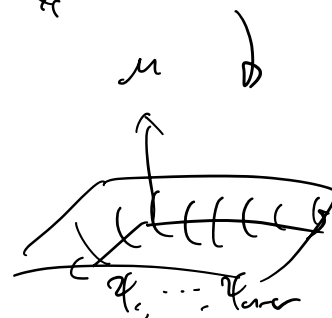
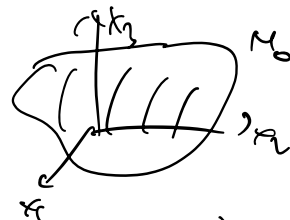
$$L_f^r h(x) + L_g L_f^{n-r} h(x) u(x)$$

M_0

$$\parallel y^{(r)} = 0 \rightarrow u(x) = u_0(x) = - \frac{L_f^r h(x)}{L_g L_f^{n-r} h(x)}$$

+ $\mu(x)=0$ initial state

$$\Rightarrow \begin{cases} \dot{\mu} = 0 \\ \ddot{\varphi} = w(0, \varphi) \leadsto 0\text{-dynamics} \end{cases}$$



$$u_0(\varphi) = - \frac{a(0, \varphi)}{b(0, \varphi)}$$

ex 6.12

$$\dot{x} = \begin{pmatrix} -x_1 \\ 2x_1x_2 + x_2^2 \\ 2x_2 \end{pmatrix} + \begin{pmatrix} e^{2x_2} \\ \frac{1}{2} \\ 0 \end{pmatrix} u$$

$$y = h(x) = x_3.$$

$$-1) (\mu_1, \mu_2, \varphi) = (x_3, 2x_2, (1+x_1)e^{2x_2}) \equiv \text{normal coordinates}$$

$$\left. \begin{aligned} \mu_1' &= \mu_2 \\ \mu_2' &= 2(1+\varphi + e^{\mu_2})\mu_2 + 2s_1(\mu_2/2) + u \\ \varphi' &= (-\varphi - e^{\mu_2})(1 + 2\mu_2 e^{\mu_2}) - 2s_2(\mu_2/2)e^{\mu_2} \end{aligned} \right\} \text{normal form.}$$

$$\text{0-dynamics: let } \mu_1 = 0, \mu_2 = 0$$

$$\rightarrow \varphi^0 = -\varphi, \quad u_0 \stackrel{\mu_2^0=0}{=} 0$$

$$\text{Def } \begin{cases} \dot{x} = f(x) + g(x)u \\ y = h(x) \end{cases}$$

asymptotically minimum phase

if its 0-dynamics : asymptotically stable

globally asymptotically stable minimum phase $\Leftrightarrow$ 0-dynamics : asymptotically stable

for any $z(0)$

Local Asymptotic Stabilization

0-dynamics = local stability!

relative degree r .

$$y^{(r)} = L_f^r h(x) + L_g L_f^{r-1} h(x) u$$

$$u = \frac{1}{L_g L_f^{r-1} h(x)} (-L_f^r h(x) + v)$$

$$-1 y^{(r)} = v$$

$$\text{Put } v = -k_{r-1} y^{(r-1)} - \dots - k_1 \dot{y} - k_0 y$$

k_i chosen so that $K(p) = p^{r-1} + k_{r-1} p^{r-2} + \dots + k_1 p + k_0$ has all roots in left complex plane.

$$\Rightarrow u(x) = \frac{1}{L_g L_f^{r-1} h(x)} [-L_f^r h(x) - k_{r-1} y^{(r-1)} - \dots - k_1 \dot{y} - k_0 y]$$

Thm system rel deg = r , 0-dynamics : locally asymptotically stable.

k_i chosen so that $\sim$

$\Rightarrow$ a control law u yields a locally asymptotically stable closed loop system.

$$\text{PF)} \quad \tilde{M} = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ & & \ddots & & \\ 0 & 0 & & & 1 \\ -k_0 & -k_1 & \dots & & -k_{r-1} \end{pmatrix} \mu = A \mu$$

$$\dot{x} = w(x, u) = A_1 x + A_2 u + \text{h.o.t.} \quad \text{in Taylor expansion about } x_0 = 0$$

$$\rightarrow \frac{d}{dt} \begin{pmatrix} \mu \\ x \end{pmatrix} = \begin{pmatrix} A & 0 \\ A_1 & A_2 \end{pmatrix} \begin{pmatrix} \mu \\ x \end{pmatrix} + \text{h.o.t.}$$

0-dyn : asymp stable $\Rightarrow \dot{x} = A_2 x$ either stable or marginally stable
by Lyapunov's linearization thm. (thm 3.1)
(eigenvalues in left plane)

If A_2 : asymp stable, linearized of the whole system : asymptotically stable

$\Rightarrow$ by Lyap lin thm, nonlinear system : locally asymp stable.

If A_1 : marginally stable, stability decided by center-manifold theory.
[Byrnes and Isidori, 1988] $\square$

$$\text{ex} \quad \begin{cases} \dot{x}_1 = x_1^2 x_2 \\ \dot{x}_2 = 3x_2 + u \end{cases}$$

$$\rightarrow \text{linearization at } x=0 : \begin{cases} \dot{x}_1 = 0 \\ \dot{x}_2 = 3x_2 + u \end{cases}$$

$$\text{Define } \check{y} = -2x_1 - x_2 \quad \rightarrow \text{rel deg} = 1, \quad \frac{dy}{dt} = -2x_1^2 x_2 - 3x_2 - u$$

$$0\text{-dynamics } (y=0) : \dot{x}_1 = -2x_1^3 \quad \text{ex 3.9} \Rightarrow \text{asymp stable}$$

$$u = -2x_1^2 x_2 - 4(x_2 - 2x_1) \quad \text{locally stabilizes the system by thm 6.3.}$$

$$\left(u(x) = \frac{1}{\underbrace{L_{\check{f}}^{r-1} \check{g}}_{-1}} \left[\underbrace{-L_{\check{f}}^r \check{g} - k_{r-1} \check{g}^{(r-1)} - \dots - k_0 \check{g}}_{\substack{1 - 2x_1 - x_2 \\ \rightarrow p(s)=0 \rightarrow s=-1}} \right] \right)$$

Global Asymptotic Stabilization

u : "input" to the internal dynamics

y : "output"

① Find a "control law" $u_0 = u_0(y)$ stabilizing the internal dynamics,

Lyapunov function V_0 demonstrating the stabilizing property.

② Lyapunov fn candidate U as a modified version of V_0 ,

choose control input v so that U be a Lyap fn for the whole dynamics.

$$\begin{cases} \dot{y} = v \\ \ddot{z} + \dot{z}^3 + yz = 0. \end{cases} \quad y = z^2 \quad \text{interval dynamics : asymp stable}$$

Lyap fn

$$V_0 = \frac{1}{2} \dot{z}^2 + \frac{1}{4} z^4$$

$$\rightarrow \dot{V}_0 = -\dot{z}^4 - z\dot{z}(y - z^2)$$

$$V = V_0 + \frac{1}{2}(y - z^2)^2 \quad \text{Lyap fn candidate}$$

$$\rightarrow \dot{V} = -\dot{z}^4 + (y - z^2)(2 - 3z\dot{z})$$

$$v = -y + z^2 + 3z\dot{z}$$

$$\rightarrow \dot{V} = -\dot{z}^4 - (y - z^2)^2$$

Invariant set thm $\Rightarrow$ the whole system state to 0.
(thm 3.5)

$$\dot{y} = v$$

$$\ddot{z} + \dot{z}^3 - 2\dot{z}^5 + yz = 0$$

$$\psi = (z \quad \dot{z})^T$$

0-dynamics unstable $\rightarrow$ non-minimum phase.

interval dynamics : asymp stable if $y = 2z^4$

$$\text{let } V_0 = \frac{1}{2} \dot{z}^2 + \frac{1}{6} z^6$$

$$\dot{V}_0 = -\dot{z}^4 - z\dot{z}(y - 2z^4)$$

$$V = V_0 + \frac{1}{2}(y - 2z^4)^2$$

$$\rightarrow \dot{V} = -\dot{z}^4 + (y-2z^4)(v - \delta z^3 \dot{z} - z \ddot{z})$$

$$v = -y + 2z^4 + \delta z^3 \dot{z} + z \ddot{z}$$

$$\rightarrow \dot{V} = -\dot{z}^4 - (y-2z^4)^2$$

$\Rightarrow$ 1 whole system state to zero.

$$\begin{cases} \ddot{x} + x^2 y z e^{xz} = (x^4 + 2)u \\ \ddot{y} + y^3 z^2 - x = 0 \\ \ddot{z} + z^3 - z^5 + yz = 0 \end{cases}$$

$$v = (x^4 + 2)u - x^2 y^5 z e^{xz}$$

$$\rightarrow \begin{cases} \ddot{x} = v \\ \ddot{y} + y^3 z^2 - x = 0 \\ \ddot{z} + z^3 - z^5 + yz = 0 \end{cases}$$

$$\text{control law } x = x_0 = y^3 z^2 - y + 2z^4 + \delta z^3 \dot{z} + z \ddot{z}$$

$\rightarrow$ globally asymp stabilize y, z .

$$\text{using } V_0 = \frac{1}{2} \dot{z}^2 + \frac{1}{6} z^6 + \frac{1}{2} (y-2z^4)^2$$

$$V = V_0 + \frac{1}{2} (x-x_0)^2$$

$$\rightarrow \dot{V} = -\dot{z}^4 - (y-2z^4)^2 + (x-x_0)(y-2z^4 + v - \dot{x}_0)$$

$$v = \dot{x}_0 - y + 2z^4 - x + x_0$$

$$-1 \quad \dot{y} = -x^2 - (y - 2x^2)^2 - (x - x_0)^2$$

$$u = \frac{v + x^2 y^5 x e^{xy}}{x^4 + 2}$$

$$\begin{cases} \dot{x} = f(x) + g(x)u \\ y = h(x) \end{cases} \quad \text{Tracking Control}$$

track $y_d(t)$.

$$\mu_d := [y_d \quad \dots \quad y_d^{(n-1)}]^T$$

$$\tilde{\mu}_d := \mu_d - \mu_d^{\text{ref}}$$

Thm 6.4. Let $\deg v_r$, $\mu_d : \mathbb{C}^\infty$, bdd,

$\varphi_d^0 = \omega(\mu_d, \varphi_d)$, $\varphi_d(0) = 0$, exists, bdd, unif asymp stable

$$u = \frac{1}{L_g L_f^{n-r} \mu_1} [-L_f^n \mu_1 + y_d^{(n)} - k_{n-1} \tilde{\mu}_n - \dots - k_0 \tilde{\mu}_1]$$

$\Rightarrow$ whole system remains bdd, $\tilde{\mu} \rightarrow 0$ exponentially.

pf) [Isidori, 1989]